Gym Tracker — MPD

Notation relationnelle complete + LDD (SQL)

1. Notation relationnelle

USER
(**id**, name, email, password)

WORKOUT_DAY
(**id**, name)

MUSCLE_GROUP
(**id**, name)

EXERCISE
(**id**, name, *muscle_group_id*)

WORKOUT
(**id**, *user_id*, *workout_day_id*, date)

WORKOUT_DAY_EXERCISE
(**id**, *workout_day_id*, *exercise_id*)

WORKOUT_EXERCISE_RESULT
(**id**, *workout_id*, *exercise_id*, sets, reps, weight)

Legende : **attribut souligne + gras** = cle primaire (PK) *attribut italique rouge* = cle etrangere (FK)

2. Cles etrangeres et cardinalites

WORKOUT.user_id	--> USER.id	N : 1
WORKOUT.workout_day_id	--> WORKOUT_DAY.id	N : 1
EXERCISE.muscle_group_id	--> MUSCLE_GROUP.id	N : 1
WORKOUT_DAY_EXERCISE.workout_day_id	--> WORKOUT_DAY.id	N : 1
WORKOUT_DAY_EXERCISE.exercise_id	--> EXERCISE.id	N : 1
WORKOUT_EXERCISE_RESULT.workout_id	--> WORKOUT.id	N : 1
WORKOUT_EXERCISE_RESULT.exercise_id	--> EXERCISE.id	N : 1

WORKOUT_DAY <-> EXERCISE : relation N:N via table d'association WORKOUT_DAY_EXERCISE

3. LDD — Script SQL (CREATE TABLE)

```
CREATE TABLE USER (
    id          INT             PRIMARY KEY AUTO_INCREMENT,
    name        VARCHAR(100)    NOT NULL,
    email       VARCHAR(150)    NOT NULL UNIQUE,
    password    VARCHAR(255)    NOT NULL
);
```

```
CREATE TABLE WORKOUT_DAY (
    id          INT             PRIMARY KEY AUTO_INCREMENT,
    name        VARCHAR(100)    NOT NULL
);
```

```
CREATE TABLE MUSCLE_GROUP (
    id          INT             PRIMARY KEY AUTO_INCREMENT,
    name        VARCHAR(100)    NOT NULL
);
```

```
CREATE TABLE EXERCISE (
    id                  INT             PRIMARY KEY AUTO_INCREMENT,
    name                VARCHAR(150)    NOT NULL,
    muscle_group_id     INT             NOT NULL,
    CONSTRAINT fk_exercise_muscle
        FOREIGN KEY (muscle_group_id) REFERENCES MUSCLE_GROUP(id)
);
```

```
CREATE TABLE WORKOUT (
    id                  INT             PRIMARY KEY AUTO_INCREMENT,
    user_id             INT             NOT NULL,
    workout_day_id      INT             NOT NULL,
    date                DATE            NOT NULL,
    CONSTRAINT fk_workout_user
        FOREIGN KEY (user_id) REFERENCES USER(id),
    CONSTRAINT fk_workout_day
        FOREIGN KEY (workout_day_id) REFERENCES WORKOUT_DAY(id)
);
```

```
CREATE TABLE WORKOUT_DAY_EXERCISE (
    id                  INT             PRIMARY KEY AUTO_INCREMENT,
    workout_day_id      INT             NOT NULL,
    exercise_id         INT             NOT NULL,
    CONSTRAINT fk_wde_day
        FOREIGN KEY (workout_day_id) REFERENCES WORKOUT_DAY(id),
    CONSTRAINT fk_wde_exercise
        FOREIGN KEY (exercise_id) REFERENCES EXERCISE(id),
    CONSTRAINT uq_day_exercise
        UNIQUE (workout_day_id, exercise_id)
);
```

```
CREATE TABLE WORKOUT_EXERCISE_RESULT (
    id                  INT             PRIMARY KEY AUTO_INCREMENT,
    workout_id          INT             NOT NULL,
    exercise_id         INT             NOT NULL,
    sets                INT             NOT NULL,
    reps                INT             NOT NULL,
    weight              DECIMAL(6,2)    NOT NULL,
    CONSTRAINT fk_wer_workout
        FOREIGN KEY (workout_id) REFERENCES WORKOUT(id),
    CONSTRAINT fk_wer_exercise
        FOREIGN KEY (exercise_id) REFERENCES EXERCISE(id)
);
```

Types choisis a titre indicatif (MySQL / MariaDB). Adapter selon le SGBD cible (PostgreSQL, H2, etc.).
PK = Primary Key · FK = Foreign Key · AUTO_INCREMENT peut devenir SERIAL / IDENTITY selon le moteur.